



CVR COLLEGE OF ENGINEERING

Department of CSE(Cyber Security)

B.Tech CSE(CS) IV Year I Semester

NGO Management System Using Blockchain

Project Stage-I, Review – I
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Under the guidance of
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Domain Introduction

- **Domain:** Cyber Security & Blockchain

Introduction:

NGOs manage large amounts of donations, campaigns, expenditures, and documents, but ensuring transparency, accountability, and donor trust in how funds are utilized remains a challenge. Donors may not have an easy way to track their contributions, and financial records can be difficult to verify.

To address these issues, our NGO Management System provides a centralized platform where the Admin, NGOs, and Donors can manage and monitor NGO activities. Donations and expenditures can be securely recorded and verified, allowing donors to track the usage of funds and NGOs to maintain reliable records. This helps create a more transparent, accountable, and trustworthy NGO management process.

Significance:

- Improves transparency in NGO donations and expenditures.
- Provides accountability by maintaining verifiable financial records.
- Builds donor trust by enabling donation tracking and verification.
- Simplifies NGO operations through centralized management of activities and records.

Description of Challenges and problem statement

Current Challenges:

1. **Lack of Transparency** – Donors often have limited visibility into how their donations are received and utilized.
2. **Risk of Data Tampering** – Centralized databases allow authorized users to modify financial records, making it difficult to guarantee data integrity.
3. **Poor Donation and Expenditure Traceability** – Tracking the complete flow of funds from donation to expenditure can be difficult and time-consuming.
4. **Manual Financial Document Management** – Managing and verifying invoices, receipts, and other financial documents manually can lead to errors and inefficiency.
5. **Limited Financial Analysis and Accessibility** – Existing systems often lack intelligent financial analysis and natural-language querying, making it difficult for users to quickly obtain meaningful financial insights.

Problem Statement:

The main problem is the lack of transparency, traceability, and tamper-resistant financial records in conventional NGO management systems. Centralized databases may allow modification of records, while manual document processing and limited financial analysis make monitoring donations and expenditures difficult. Our project addresses these challenges by combining blockchain for immutable transaction records with secure NGO management, financial analytics, document processing, and an AI-powered financial assistant.

Description of Existing Methodologies

S. No.	Existing Methodology	How It Works	Advantages	Disadvantages / Drawbacks	Citation (Post-2024 Paper)
1	Traditional Centralized NGO Management	Stores NGO, donor, donation, and expenditure information in a centralized database controlled by the organization.	Simple implementation; easy database management; low initial cost.	Single point of control; records can be modified; limited independent verification and transparency.	Mariyam et al., 2025
2	Centralized Online Donation Platforms	Donors contribute through a centralized website, while the platform maintains donation and campaign records.	Easy for donors to use; convenient online donations; centralized reporting.	Donors have limited visibility after donation; depends heavily on platform/NGO records.	Sung et al., 2023
3	Manual Donation & Expenditure Tracking	NGOs maintain donation, expenditure, bills, and beneficiary records using spreadsheets or manual documentation.	Easy to implement; familiar to NGO staff; no blockchain infrastructure required.	Time-consuming; prone to human error; difficult to audit; records can be altered or lost.	Palma, 2024

Description of Existing Methodologies

S. No.	Existing Methodology	How It Works	Advantages	Disadvantages / Drawbacks	Citation (Post-2024 Paper)
4	Centralized Cloud-Based Record Management	NGO financial records and documents are stored on centralized cloud/database servers and accessed through web applications.	Remote access; scalable storage; convenient document management.	Centralized trust; unauthorized modification remains possible; integrity depends on server controls.	Palma, 2024
5	Blockchain Donation Traceability	Each donation is recorded as a blockchain transaction, allowing stakeholders to trace funds from donor to recipient.	Immutable records; improved traceability; transparent transaction history.	Blockchain transaction costs; scalability issues; mainly focuses on donation tracking.	Almaghrabi & Alhogail, 2022
6	Smart Contract-Based Charity Management	Smart contracts automatically record and manage donations according to predefined rules on a blockchain..	Automation; reduces intermediaries; improves accountability and transparency.	Smart-contract bugs can cause security problems; requires blockchain knowledge.	Nairi et al., 2024

Description of Existing Methodologies

S. No.	Existing Methodology	How It Works	Advantages	Disadvantages / Drawbacks	Citation (Post-2024 Paper)
7	Blockchain-Based Donation System with Trust Verification	Blockchain provides transparent and immutable donation information that donors can use to evaluate an NGO's activities.	Improves donor trust; provides transparency and immutability; enables donation tracking.	Does not necessarily provide complete NGO administration or expenditure-document management.	Sung et al., 2023
8	Blockchain-Based Philanthropic System with Authentication	Uses blockchain, cryptographic hashing, smart contracts, and authentication mechanisms to secure charity transactions.	Better authentication; tamper-resistant records; improved privacy and traceability.	More complex implementation; authentication and blockchain integration increase system complexity.	Ahmed et al., 2024
9	Blockchain-Based Charity Platform with Hash Verification	Donation records are assigned unique hashes and stored on blockchain so that modifications can be detected.	Tamper detection; verifiable transactions; improved donor confidence.	Primarily focuses on donation transactions; limited expenditure and document management.	Narasimhulu et al., 2025
10	Ethereum-Based Transparent Charity Framework	Uses Ethereum smart contracts and distributed ledger technology to track donations from contribution to final allocation.	Immutable records; real-time tracking; fraud reduction; improved accountability.	Public blockchain dependency; scalability and transaction-cost concerns; limited complete NGO management.	Mariyam et al., 2025

Description of Proposed Solution

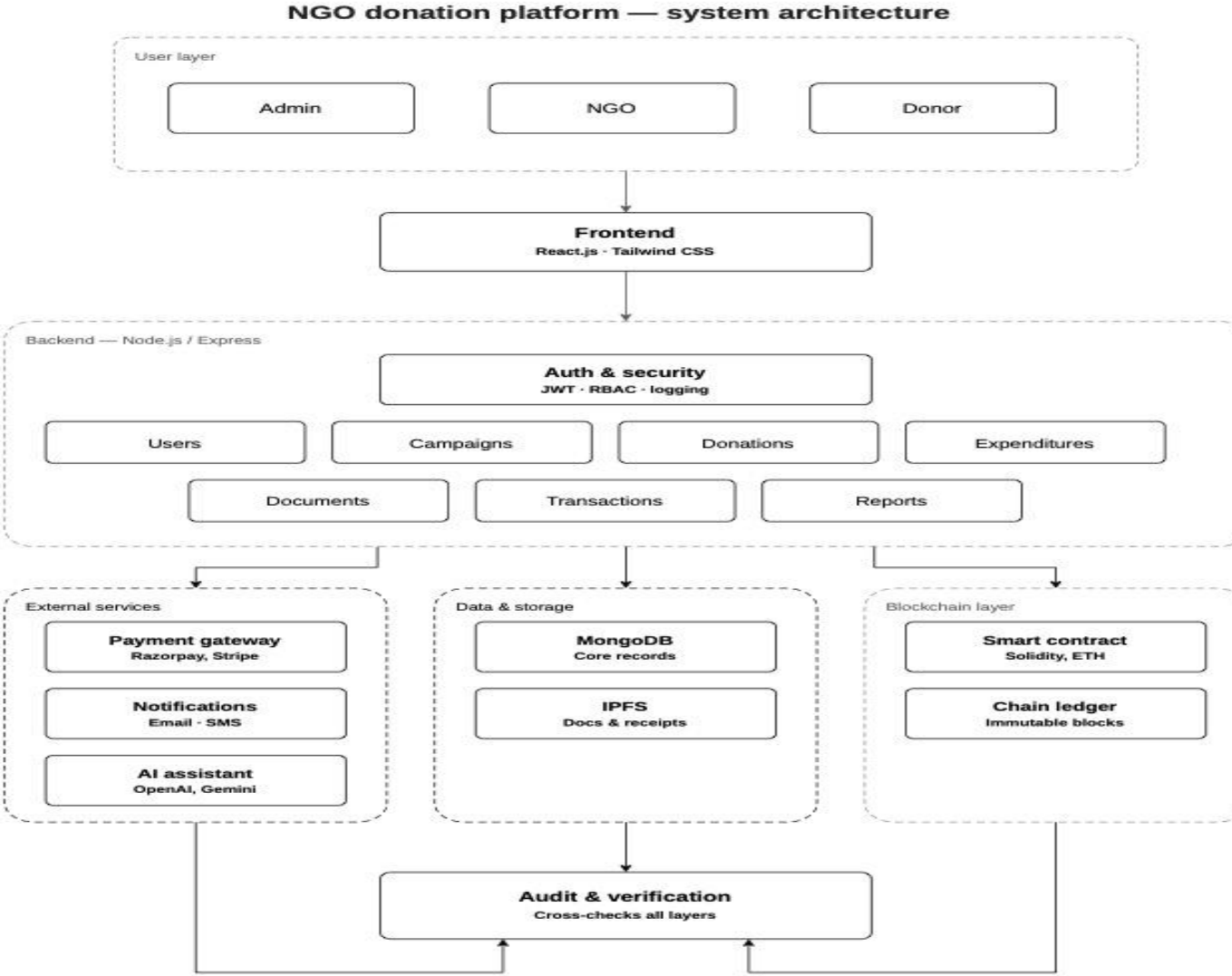
The proposed **NGO Management System using Blockchain** is a secure and transparent web-based platform designed to manage NGO donations and expenditures efficiently. The system uses **blockchain technology and smart contracts** to maintain tamper-resistant records of important financial transactions, improving transparency and accountability.

- **Blockchain-Based Records:** Donations and expenditures are recorded on the blockchain through smart contracts, providing an immutable and traceable transaction history.
- **Donation and Expenditure Management:** NGOs can record, monitor, and track incoming donations and outgoing expenditures, allowing the complete flow of funds to be monitored.
- **AI-Powered Financial Assistant:** An AI module allows users to ask financial questions in **natural language**, such as total donations, expenditures for a particular period, or category-wise spending, and receive meaningful responses.
- **Document and Financial Analysis:** The system supports invoice and receipt management and can use AI to extract and summarize relevant information, while dashboards provide graphical insights into donations and expenditures.

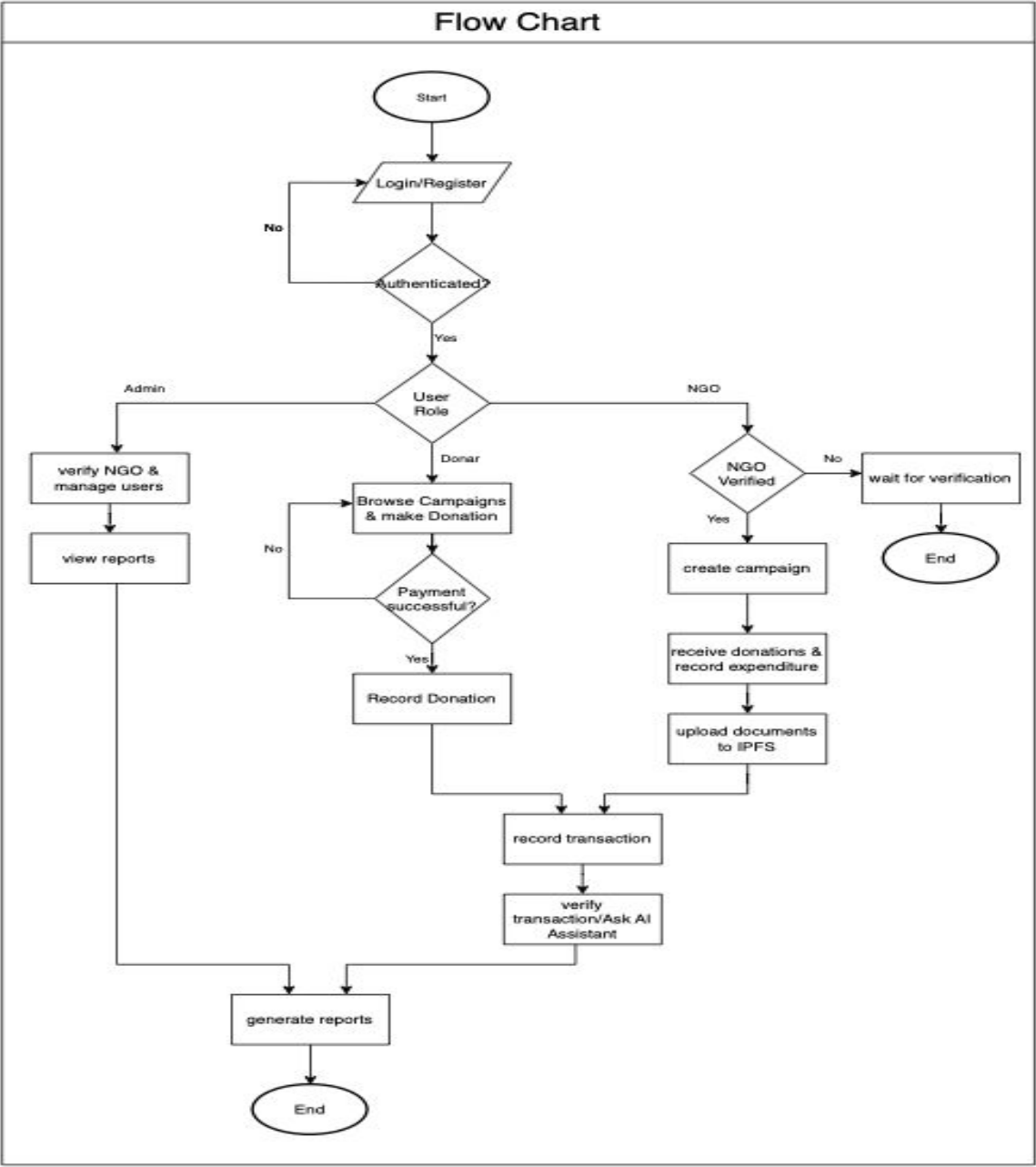
System Design & Diagrams:

- **Trust without a central authority.** Donors don't have to take an NGO's word for how funds were spent every donation and expenditure is written to an immutable ledger that anyone can independently verify, removing the need to trust the NGO's internal bookkeeping alone.
- **Tamper-evident audit trail.** Once a transaction is recorded on-chain, it cannot be altered or deleted, so fraud or misreporting (e.g. inflating donation totals, hiding an expense) becomes detectable rather than something an auditor has to take on faith.
- **End-to-end traceability of funds.** Because donations and expenditures are both linked to the same campaign ID on-chain, a donor can trace their specific contribution from the moment it's given through to the receipt of what it was spent on — something a traditional centralized database can claim but not prove.

System Architecture:



Flow Chart:



Description of Proposed Solution

EXPECTED OUTCOMES

Functional / System Outcomes

Working blockchain-based NGO transparency prototype implementing a two-stage flow:

- 1. Stage-I: Donation & Expenditure Recording** — donor initiates payment → backend validates via payment gateway → smart contract records the transaction → document proof hashed to IPFS → immutable ledger entry created.
- 2. Stage-II: Continuous Verification & Reporting** — every on-chain transaction is independently verifiable by any donor → AI Assistant analyzes spending patterns and flags anomalies → real-time transparency dashboard for Admin, NGO, and Donor roles.

NOVELTY OVER EXISTING METHODS

Feature	Traditional VPN	1st-Gen ZTNA	Periodic Re-auth Schemes	Proposed Framework
Trust model	Manual paper-based, trust the NGO	Payment confirmed, spending untracked	On-chain Donations Only	Donation to expenditure traceability
Transparency	Annual audited report only	None after payment	Donation-side only	End-to-End: donation, spend and receipt all visible
Verification	Manual Audit, delayed by months	None	on-chain but no proof of what money bought	on-chain + IPFS hashed receipts verifiable anytime

Hardware and Software Requirements

Hardware Specifications

- CPU RAM: 16 GB
- Processor: Intel Core i5/i7, 11th Gen or above
- Storage: 512GB SSD

Software Specifications

- Operating System: Windows 10/11
- Programming Languages: JavaScript, Solidity, Python
- Frontend: React.js, HTML, CSS
- Backend: Node.js, Express.js
- Blockchain: Ethereum, Hardhat, Solidity
- Blockchain Interaction: Ethers.js
- Database Management: MongoDB
- AI/NLP: Python, NLP/LLM APIs
- Development Environment: VS Code

Timeline for next review

TASK ID	TASK NAME	START DATE	END DATE	DURATION in days	WEEKS													
					1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	Research and Analysis																	
2	Requirements																	
3	Literature Review																	
4	Documentation																	
5	Review-1																	
6	Design																	
7	Documentation																	
8	Review-2																	

The Start Date and End Date Can be Modified According to Your Project

References

1. Narasimhulu et al. (2025) — Blockchain-Based Authentic Charity Platform
2. Mariyam et al. (2025) — A Blockchain-Based Framework for Enhancing Transparency and Traceability in Charity Donations
3. Migliorini et al. (2025) — A Blockchain-Based Platform for Ensuring Provenance and Traceability of Donations for Cultural Heritage
4. Jadhav et al. (2025) — Block Chain Technology-Based Crypto-Go-Charity for Scholarship Donation Tracking Platform
5. Nairi et al. (2024) — Smart Blockchain Networks: Revolutionizing Donation Tracking in the Web 3.0
6. Ahmed et al. (2024) — Blockchain-Empowered Decentralized Philanthropic Charity for Social Good
7. Zhou, Lei & Bao (2024) — EISDSPA: An Efficient and Secure Blockchain-Based Donation Scheme with Privacy Protection and Auditability
8. Zhang et al. (2023) — Design and Implementation of Donation Information Traceability System Based on Blockchain
9. Govindarajan et al. (2023) — Blockchain Fundraising and Charity Platform
10. Sung et al. (2023) — Effect of Blockchain-Based Donation System on Trustworthiness of NPOs
11. Almaghrabi & Alhogail (2022) — Blockchain-Based Donations Traceability Framework
12. Saraswat et al. (2022) — UpHaaR: Blockchain-Based Charity Donation Scheme to Handle Financial Irregularities